

Meddies-PII: A Multilingual Framework for Personally Identifiable Information Extraction in Clinical De-identification

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Abstract

Clinical de-identification depends on accurately identifying personally identifiable information (PII). However, manually annotated datasets are costly to build, and existing synthetic alternatives often provide limited generation details or use relatively simple synthesis strategies. We introduce Meddies-PII-Dataset, which contains one million synthetic clinical documents across seventeen languages and nine PII labels. Attribute-conditioned prompts generate the documents, and thirteen deterministic gates check their structural and annotation validity. To assess the dataset's utility, we train Meddies-PII-Model, a BIOES token classifier. We compare it with existing PII extraction systems using exact-match entity-level F1. Meddies-PII-Model achieves the highest score among the evaluated systems on all reported benchmarks, with a mean F1 of 0.827 across fifteen external benchmarks compared with 0.658 for the strongest baseline. Upon acceptance, we will publicly release the dataset, benchmark, model, generation framework, and evaluation code to support research on multilingual clinical de-identification.

Keywords: Natural Language Processing; Named Entity Recognition; Personally Identifiable Information; Synthetic Data; Healthcare

1 Introduction

Named Entity Recognition (NER) is a fundamental task in natural language processing. In privacy-sensitive domains such as healthcare, however, the primary objective is identifying personally identifiable information (PII), a specialized subset of named entities whose disclosure may compromise patient privacy (Savkin et al., 2025). Although PII extraction is commonly formulated as an NER task, it differs from general-purpose NER in both its ontology and application objectives. Consequently, data-driven PII systems require dedicated

datasets rather than relying solely on conventional NER corpora. Existing PII datasets are predominantly manually annotated, making them expensive to construct and limiting their scale, language coverage, and domain diversity (Neamatullah et al., 2008; Uzuner et al., 2007; Stubbs et al., 2015). Although several synthetic PII datasets have recently been proposed, most either do not disclose their generation pipeline or rely on relatively simple prompting strategies (Ai4Privacy, 2026; AI, 2024; Steier et al., 2025).

We propose a multilingual synthetic data generation framework that combines attribute-conditioned prompting with deterministic validation to construct Meddies-PII-Dataset. The framework controls language, document type, text format, scenario, ontology constraints, and challenging edge cases to generate diverse and structurally complex documents. Generated samples that violate the declared requirements are repaired or rejected before inclusion in the dataset. We further train a token-classification model to evaluate the utility of the resulting data. We evaluate the model on fifteen external benchmarks and the Meddies-PII Benchmark. Our benchmark combines standard and challenging examples under the Meddies-PII ontology to assess performance across different levels of difficulty.

This paper makes three contributions:

1. We introduce **Meddies-PII-Dataset**, a synthetic clinical PII dataset containing one million documents across seventeen languages and annotated with a nine-label PII ontology.
2. We present a controlled multilingual synthesis framework that combines multiple document and generation attributes with deterministic structural and annotation validation.
3. We introduce **Meddies-PII-Model**, a BIOES token classifier, and show that it achieves the

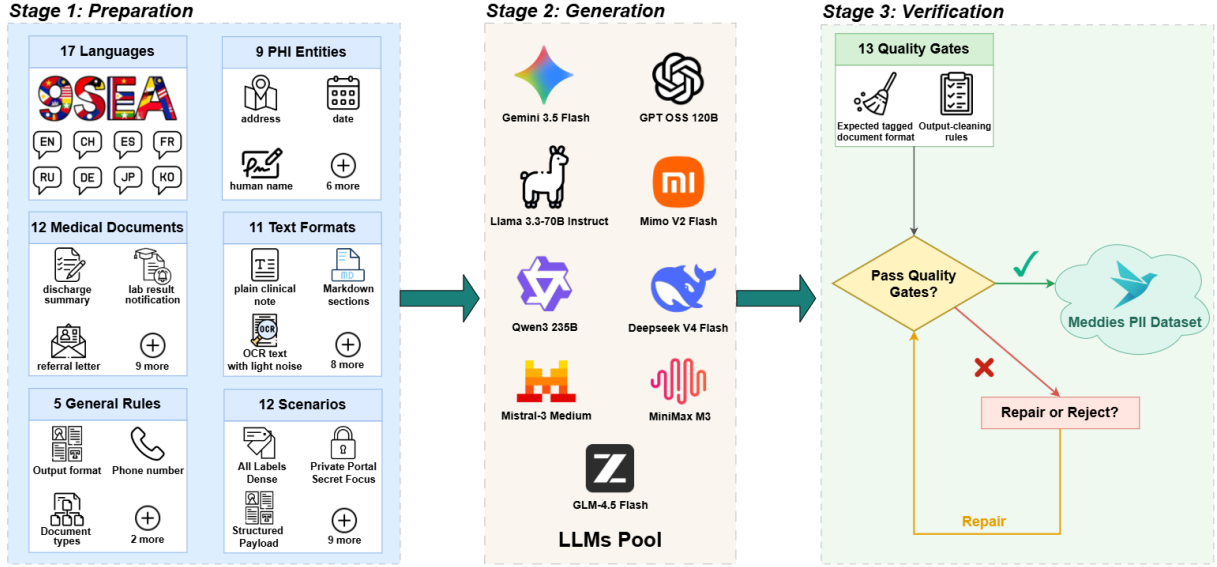


Figure 1: Synthetic data generation and verification pipeline to create **Meddies-PII-Dataset**.

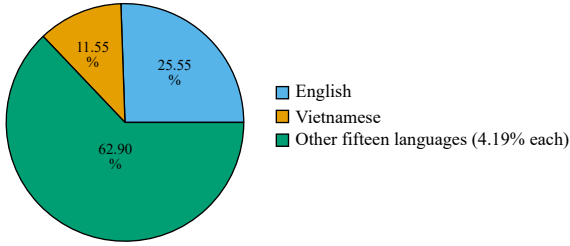


Figure 2: Language distribution of Meddies-PII-Dataset. The other fifteen languages each contain 41,934 samples (4.19%).

We then refined the generation process and created a second collection of higher-quality samples. The final dataset mixes selected samples from both collections to reach one million documents while preserving the target language distribution and broader coverage. This design is consistent with sample-wise pretraining data mixing research that jointly considers quality and diversity when determining the resulting data distribution (Xi et al., 2025). The language selection emphasizes Southeast Asian languages, several of which have limited representation in existing PII datasets. Because widely spoken international languages are also used across Southeast Asian countries, we extend the dataset with prominent languages beyond the region, yielding coverage across different scripts, language families, and resource levels. Figure 2 summarizes the language distribution.

PII labels We define a nine-label PII ontology informed by the identifiers and distinguishing at-

tributes described in ISO/IEC 29100:2024 (International Organization for Standardization and International Electrotechnical Commission, 2024):

- **address**: street addresses, postal codes, coordinates, and care locations;
- **company_name**: hospitals, clinics, departments, companies, and other organizations;
- **date**: birth dates, admission and discharge dates, and other specific datetimes linked to a person or encounter;
- **email_address**: email addresses;
- **human_name**: patient, clinician, and other person names;
- **id_number**: medical record numbers, government identifiers, passport and licence numbers, IP addresses, and account numbers;
- **phone_number**: telephone and fax numbers;
- **private_url**: access-bearing patient-portal, result, signed, or private-record URLs;
- **secret**: passwords, API keys, authentication or session tokens, cookies, personal identification numbers, and one-time passwords.

Meddies-PII Benchmark Our benchmark contains 5,100 validated documents distributed evenly across the seventeen languages, with 300 documents per language. For each language, 100 documents assess PII extraction under standard conditions, while 200 emphasize adversarial surface forms. Each adversarial document combines four to five of thirteen perturbation techniques. These include spoken digits and email symbols, phonetic spelling, unusual spacing, entity-internal line breaks, truncation, partial masking, OCR-like substitutions, Unicode confusables, and concatenated values. These perturbations preserve the annotated PII spans while testing robustness to noise commonly encountered in speech transcripts, scanned records, messages, and administrative documents.

3.2 Dataset Creation

As shown in Figure 1, dataset creation proceeds through three stages: Preparation, Generation, and Verification. For each sample, the framework prepares a controlled prompt, sends it to one model from the generation pool, and validates the resulting document before accepting it into the dataset.

Stage 1: Preparation Each generation prompt combines six attribute groups that control the content and structure of a sample. Medical documents determine the workflow and content, while text formats determine how that content is expressed. General rules maintain annotation consistency, and scenarios control the operational context and difficulty. Together, these groups capture complementary sources of variation in clinical PII data. Randomly combining attribute values produces varied prompts at scale, an approach supported by evidence that attributed prompts improve synthetic-data diversity and downstream performance while reducing query cost (Yu et al., 2023). We refined the final set of options through project discussions to balance realistic clinical and administrative use cases with broad label and structural coverage. Figure 3 shows the prompt template.

1. **Languages:** Specifies the target language of the generated document.
2. **PII labels:** Nine labels specify the PII categories and required label coverage for each document.
3. **Medical documents:** Represents PII in common clinical and administrative workflows.

The twelve documents are discharge summary; outpatient note; lab-result notification; patient portal access log; insurance claim note; referral letter; pharmacy refill request; appointment reminder; telehealth triage note; medical bill or invoice; FHIR-like Patient and Encounter resource; and HL7-style admission or update message.

4. **Text formats:** Captures how equivalent information changes across prose, structured records, logs, transcripts, and noisy text. The eleven formats are plain clinical note; Markdown sections; messy nurse or administrative note; table-like rows; JSON-like object; FHIR-like JSON; HL7-like pipe-delimited text; patient portal audit log; voice-scribe transcript; lightly degraded OCR text; and clinical dialogue transcript.
5. **General rules:** Maintains consistent generation and annotation across combinations of attributes. The five rule groups govern the output format, phone-number construction, private URLs, secrets, and compatibility between medical documents and text formats.
6. **Scenarios:** Places PII within realistic operational contexts while controlling label density and difficulty. The twelve scenarios are All Labels Dense; Private Portal Secret Focus; Adversarial Obfuscation; Public Negative Contrast; Structured Payload; General Consumer Admin Support; HR, Employer, School, Insurance, or Telecom Admin; Code, Log, Security, or Account Recovery; Adversarial Formatting; Structured General Payload; One-Hop Deferred Clue; and Public Negative URL and Data Contrast.

```

user_prompt = f"""
Generate one synthetic {language} {document_type}
in {text_format} format.

Scenario: {scenario_name}
{scenario_description}

Hard constraints:
1. Use at least {min_spans} tagged PII spans.
2. Use at least {min_unique_labels} unique Meddies
   labels.
3. Required labels for this sample: {
   required_labels}.
etc.

Required-label coverage before you answer: {", ".
join(f"{label}=present" for label in
resolved_required_labels) or "none"}.
Minimum coverage before you answer: at least {
min_spans} bracketed tags and at least {
min_unique_labels} unique labels.

Remember: label syntax must be `[value]<label>`
and labels must be one of: {", ".join(f"<{label}>"
for label in PII_LABELS)}.
"""

```

Fig. 3. User-prompt template for Meddies-PII-Dataset generation. Braced fields are populated from the sampled generation attributes and validation requirements.

Hard constraints The hard-constraint block converts the sampled attributes into explicit generation requirements. It specifies the minimum number of tagged spans and unique labels, lists labels that must appear, and controls when private URLs, secrets, negative examples, and adversarial perturbations are required. It also enforces the exact `[value]<label>` syntax, complete entity boundaries, document-only output, and a final coverage self-check. These constraints guide the model toward drafts that can be checked by the deterministic gates in Stage 3, although a draft is accepted only after passing verification.

Stage 2: Generation The generation pool contains nine large language models (LLMs) from different model families. These are Gemini 3.5 Flash (Google DeepMind, 2026), GLM-4.5-Flash (Zeng et al., 2025), Llama 3.3 70B Instruct (Meta AI, 2024), MiMo-V2-Flash (Xiao et al., 2026), Mistral Medium 3 (Mistral AI, 2025), gpt-oss-120b (Agarwal et al., 2025), DeepSeek-V4-Flash (DeepSeek-AI, 2026), Qwen3-235B-A22B (Yang et al., 2025), and MiniMax M3 (MiniMax, 2026). For each sample, the framework sends the assembled prompt to one selected LLM from this pool. Distributing generation across model families reduces the pipeline’s dependence on the behavior of a single generator.

Stage 3: Verification Each generated draft enters deterministic verification. Verification uses regular expressions, a deterministic character-level parser, rule-based checks, and SHA-256 hashing. Regular expressions check annotation syntax and language-specific Unicode ranges. The parser reconstructs the raw document and verifies span offsets, while rule-based checks validate phone numbers. SHA-256 hashes detect duplicate documents. Together, thirteen gates check label coverage, span count, document length, offsets, phone-number structure, native-script presence, and duplication. Drafts that pass are accepted, while failed drafts enter the repair workflow. API failures and empty content are rejected because they contain no recoverable document, while unknown scenarios are rejected because the metadata needed to interpret and validate the document is unavailable. For the remaining drafts, conservative regex repair fixes only explicit labels with unambiguous value boundaries, such as misplaced brackets, spacing around tags, or unwrapped structured-field values. Repaired drafts are then revalidated and accepted only if they pass every gate and are unique. Drafts with missing or unknown labels, too few spans, invalid lengths or offsets, templated phone numbers, or missing native script are rejected because correcting them would require inventing content, labels.

4 Evaluation

4.1 Evaluation Settings

Evaluation Protocol We compare Meddies-PII-Model with four publicly available PII extraction systems: OpenAI Privacy Filter (OpenAI, 2026), GLiNER2 (Zaratiana et al., 2025), SuperClinical-Large-434M-v1 (Science, 2026), and LFM2.5-Encoder-350M-PII-Detector (Liquid AI, 2026). The external evaluation includes twelve language subsets from OpenPII 1.5M (Ai4Privacy, 2026), together with CredData (Yun et al., 2021), Gretel PII Masking (AI, 2024), and Nemotron-PII (Steier et al., 2025). We additionally evaluate every system on the Meddies-PII Benchmark. Labels produced by each system are mapped to the shared PII ontology before scoring. We report exact-match entity-level micro-F1 by pooling entity counts across documents. A predicted entity is a true positive only when both its PII label and character boundaries exactly match a reference entity. Predictions with an incorrect label or bound-

ary are counted as false positives, while unmatched reference entities are counted as false negatives. Using the pooled counts, the metric is computed as

$$\begin{aligned} P &= \frac{TP}{TP + FP}, & R &= \frac{TP}{TP + FN}, \\ F_1 &= \frac{2PR}{P + R} = \frac{2TP}{2TP + FP + FN}. \end{aligned} \quad (1)$$

Model Development Our initial model development treated PII extraction as autoregressive structured generation. We first applied supervised fine-tuning (SFT) and subsequently reinforcement learning (RL), but sequential decoding made inference slow and the resulting extraction quality was not competitive. We therefore adopted the one-pass token-classification approach used by OpenAI Privacy Filter (OpenAI, 2026). This approach classifies each token using bidirectional document context instead of generating entities token by token. BIOES marks tokens as Beginning, Inside, End, Single-token, or Outside (O) relative to an entity. Meddies-PII-Model fine-tunes LFM2.5-350M with a classifier containing 37 output classes: O and four states for each of the nine PII labels. A linear classification head predicts a BIOES state for each token. Constrained Viterbi decoding then converts the predicted sequence into character-level entity spans while enforcing valid BIOES transitions (Lester et al., 2020).

4.2 Results

Overall comparison Table 1 reports exact-match entity-level micro-F1 for the evaluated systems. Meddies-PII-Model ranks first on every benchmark shown. Across the fifteen external benchmarks, it achieves an unweighted mean F1 of 0.827, exceeding OpenAI Privacy Filter, the strongest baseline by external mean, by 0.169. When the Meddies-PII Benchmark is included, the overall means are 0.833 and 0.644, respectively. The consistent advantage across both external and in-domain evaluations suggests that the improvement is not confined to the distribution used to develop Meddies-PII-Model.

Multilingual performance On the twelve OpenPII 1.5M language benchmarks, Meddies-PII-Model scores between 0.739 and 0.880 and ranks first in every language. Its largest margins over the strongest competing score occur for Vietnamese (0.880 versus 0.634), Chinese (0.816

versus 0.572), and Korean (0.739 versus 0.554). This pattern indicates that the external mean is not driven only by English and that the model transfers effectively across several languages and scripts. However, the lower absolute scores for Korean and Japanese show that multilingual performance is not uniform, leaving room for language-specific analysis and improvement.

Cross-dataset performance Meddies-PII-Model achieves F1 scores of 0.590 on CredData, 0.872 on Gretel, and 0.956 on Nemotron. The strongest baseline scores on the same benchmarks are 0.370, 0.729, and 0.878, giving margins of 0.220, 0.143, and 0.078, respectively. Positive margins across all three datasets provide evidence of transfer beyond a single annotation source. CredData remains substantially more difficult for every system, indicating that differences in domain, document structure, or annotation policy still present a meaningful challenge.

Meddies-PII Benchmark The Meddies-PII Benchmark combines standard examples with examples emphasizing adversarial surface forms. On this benchmark, Meddies-PII-Model achieves an F1 of 0.878, compared with 0.540 for the strongest baseline. The 0.338 margin shows that the model handles the ontology and perturbations represented in its target setting more effectively than the compared systems. Because this is an in-domain evaluation, we report and interpret it separately from the external mean rather than treating it as direct evidence of generalization.

Taken together, the results show consistent gains across languages and datasets, while the lower scores on CredData, Korean, and Japanese identify important directions for further evaluation.

Data and code availability Upon acceptance, we will publicly release Meddies-PII-Dataset, the Meddies-PII Benchmark, Meddies-PII-Model, the generation framework, and the evaluation code.

5 Limitations

Meddies-PII-Dataset is entirely synthetic and may not capture all linguistic variation, documentation practices, and annotation ambiguity found in real-world clinical records. Its deterministic gates enforce the declared structural and annotation requirements, but they do not directly measure the semantic naturalness of every generated

Table 1: Exact-match entity-level micro-F1 scores. The best result for each benchmark is shown in bold.

Benchmark	Meddies PII Model	OpenAI Privacy Filter	GLiNER2	SuperClinical Large 434M	LFM2.5 Encoder 350M PII Detector
<i>External evaluation</i>					
OpenPII 1.5M (de)	0.827	0.716	0.663	0.620	0.610
OpenPII 1.5M (en)	0.878	0.745	0.687	0.656	0.568
OpenPII 1.5M (es)	0.870	0.794	0.732	0.683	0.655
OpenPII 1.5M (fil)	0.848	0.667	0.690	0.584	0.600
OpenPII 1.5M (fr)	0.844	0.738	0.674	0.604	0.611
OpenPII 1.5M (id)	0.848	0.664	0.688	0.566	0.627
OpenPII 1.5M (ja)	0.750	0.549	0.366	0.493	0.479
OpenPII 1.5M (ko)	0.739	0.554	0.322	0.545	0.460
OpenPII 1.5M (ms)	0.839	0.640	0.692	0.547	0.587
OpenPII 1.5M (pt)	0.855	0.756	0.692	0.687	0.651
OpenPII 1.5M (vi)	0.880	0.634	0.628	0.499	0.615
OpenPII 1.5M (zh)	0.816	0.572	0.453	0.428	0.548
CredData (en)	0.590	0.370	0.357	0.150	0.263
Gretel (en)	0.872	0.697	0.588	0.715	0.729
Nemotron (en)	0.956	0.778	0.692	0.878	0.574
External mean	0.827	0.658	0.595	0.577	0.572
Meddies-PII Benchmark	0.878	0.540	0.476	0.244	0.269
Overall mean	0.833	0.644	0.581	0.538	0.536

document. In addition, the present evaluation focuses on synthetic PII benchmarks and Meddies-PII Benchmark data; evaluation on appropriately governed real clinical records remains an important direction for future work.

6 Conclusion

We presented **Meddies-PII-Dataset**, a synthetic clinical PII dataset containing one million documents across seventeen languages and nine PII labels. After reviewing the initial generation, we refined the process, generated a second collection of higher-quality samples, and mixed both collections to preserve the target distribution and broader coverage. The improved process controls multiple document and scenario attributes and selects one generator per sample from a pool of nine models. Thirteen deterministic gates then validate each sample before acceptance. **Meddies-PII-Model** achieves the highest performance among the evaluated systems across all reported external benchmarks and the **Meddies-PII Benchmark**, demonstrating the downstream utility of the generated data. By making the dataset, benchmark, model, generation framework, and evaluation code publicly available upon acceptance, we aim to support further research on multilingual clinical de-identification and synthetic PII generation.

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References

- Sandhini Agarwal, Lama Ahmad, Jason Ai, Sam Altman, Andy Applebaum, Edwin Arbus, Rahul K Arora, Yu Bai, Bowen Baker, Haiming Bao, et al. 2025. gpt-oss-120b & gpt-oss-20b model card. *arXiv preprint arXiv:2508.10925*.
- Gretel AI. 2024. Gliner models for pii detection through fine-tuning on gretel-generated synthetic documents.
- Ai4Privacy. 2026. **OpenPII 1.5M: Multilingual PII Masking Dataset with Asia Pacific Extension**. Hugging Face dataset. Accessed: 2026-07-25.
- Ibrahim Baroud, Christoph Otto, Vera Czehmann, Christine Hovhannisyan, Lisa Raithel, Sebastian Möller, and Roland Roller. 2026. Multigrascco: A multilingual anonymization benchmark with annotations of personal identifiers. *arXiv preprint arXiv:2603.08879*.
- Ona de Gibert, Aitor García-Pablos, Montse Cuadros, and Maite Melero. 2022. **Spanish datasets for sensitive entity detection in the legal domain**. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, pages 3751–3760, Marseille, France. European Language Resources Association.

- DeepSeek-AI. 2026. [DeepSeek-V4-Flash model card](#). Hugging Face model card. Accessed: 2026-08-04.
- Louise Deleger, Todd Lingren, Yizhao Ni, Megan Kaiser, Laura Stoutenborough, Keith Marsolo, Michal Kouril, Katalin Molnar, and Imre Solti. 2014. Preparing an annotated gold standard corpus to share with extramural investigators for de-identification research. *Journal of biomedical informatics*, 50:173–183.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 4171–4186, Minneapolis, Minnesota. Association for Computational Linguistics.
- Markus Eberts and Adrian Ulges. 2019. Span-based joint entity and relation extraction with transformer pre-training. *arXiv preprint arXiv:1909.07755*.
- Google DeepMind. 2026. [Gemini 3.5 flash model card](#). Model card. Accessed: 2026-08-04.
- Cyril Grouin and Aurélie Névél. 2014. De-identification of clinical notes in french: towards a protocol for reference corpus development. *Journal of biomedical informatics*, 50:151–161.
- International Organization for Standardization and International Electrotechnical Commission. 2024. Information technology—security techniques—privacy framework. International Standard ISO/IEC 29100:2024, ISO/IEC.
- Brian Lester, Daniel Pressel, Amy Hemmeter, Sagnik Ray Choudhury, and Srinivas Bangalore. 2020. [Constrained decoding for computationally efficient named entity recognition taggers](#). In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1841–1848, Online. Association for Computational Linguistics.
- Liquid AI. 2026. [LFM2.5-Encoder-350M-PII-Detector](#). Hugging Face model card. Accessed: 2026-08-04.
- Meta AI. 2024. [Llama 3.3 70b instruct model card](#). Hugging Face model card. Accessed: 2026-08-04.
- MiniMax. 2026. [MiniMax M3 model card](#). Model card. Accessed: 2026-08-04.
- Mistral AI. 2025. [Mistral medium 3 model card](#). Model card. Accessed: 2026-08-04.
- Luise Modersohn, Stefan Schulz, Christina Lohr, and Udo Hahn. 2022. Grasco—the first publicly shareable, multiply-alienated german clinical text corpus. In *German Medical Data Sciences 2022—Future Medicine: More Precise, More Integrative, More Sustainable!*, pages 66–72. IOS Press.
- Ishna Neamatullah, Margaret M Douglass, Li-Wei H Lehman, Andrew Reisner, Mauricio Villarroel, William J Long, Peter Szolovits, George B Moody, Roger G Mark, and Gari D Clifford. 2008. Automated de-identification of free-text medical records. *BMC medical informatics and decision making*, 8(1):32.
- OpenAI. 2026. Introducing openai privacy filter. <https://openai.com/index/introducing-openai-privacy-filter/>. Accessed: 2026-07-22.
- Maksim Savkin, Timur Ionov, and Vasily Konovalov. 2025. [SPY: Enhancing privacy with synthetic PII detection dataset](#). In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 4: Student Research Workshop)*, pages 236–246, Albuquerque, USA. Association for Computational Linguistics.
- OpenMed Science. 2026. [Openmed-pii-superclinical-large-434m-v1: Pii detection model](#).
- Fábio Souza, Rodrigo Nogueira, and Roberto Lotufo. 2019. Portuguese named entity recognition using BERT-CRF. *arXiv preprint arXiv:1909.10649*.
- Amy Steier, Andre Manoel, Alexa Haushalter, and Maarten Van Segbroeck. 2025. [Nemotron-pii: Synthesized data for privacy-preserving ai](#).
- Amber Stubbs, Christopher Kotfila, and Özlem Uzuner. 2015. Automated systems for the de-identification of longitudinal clinical narratives: Overview of 2014 i2b2/uthealth shared task track 1. *Journal of biomedical informatics*, 58:S11–S19.
- Özlem Uzuner, Yuan Luo, and Peter Szolovits. 2007. Evaluating the state-of-the-art in automatic de-identification. *Journal of the American Medical Informatics Association*, 14(5):550–563.
- Xiangyu Xi, Deyang Kong, Jian Yang, Jiawei Yang, Zhengyu Chen, Wei Wang, Jingang Wang, Xunliang Cai, Shikun Zhang, and Wei Ye. 2025. [SampleMix: A sample-wise pre-training data mixing strategy by coordinating data quality and diversity](#). In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 13736–13758, Suzhou, China. Association for Computational Linguistics.
- Bangjun Xiao, Bingquan Xia, Bo Yang, Bofei Gao, Bowen Shen, Chen Zhang, Chenhong He, Chiheng Lou, Fuli Luo, Gang Wang, et al. 2026. MiMo-V2-Flash technical report. *arXiv preprint arXiv:2601.02780*.
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. 2025. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*.

- Yue Yu, Yuchen Zhuang, Jieyu Zhang, Yu Meng, Alexander J Ratner, Ranjay Krishna, Jiaming Shen, and Chao Zhang. 2023. Large language model as attributed training data generator: A tale of diversity and bias. *Advances in neural information processing systems*, 36:55734–55784.
- JaeKu Yun, ShinHyung Choi, YuJeong Lee, Oleksandra Sokol, WooChul Shim, Arkadiy Melkonyan, and Dmytro Kuzmenko. 2021. Project creddata: A dataset of credentials for research. <https://github.com/Samsung/CredData>.
- Urchade Zaratiana, Gil Pasternak, Oliver Boyd, George Hurn-Maloney, and Ash Lewis. 2025. [Gliner2: An efficient multi-task information extraction system with schema-driven interface](#). *Preprint*, arXiv:2507.18546.
- Urchade Zaratiana, Nadi Tomeh, Pierre Holat, and Thierry Charnois. 2023. Gliner: Generalist model for named entity recognition using bidirectional transformer. *arXiv preprint arXiv:2311.08526*.
- Aohan Zeng, Xin Lv, Qinkai Zheng, Zhenyu Hou, Bin Chen, Chengxing Xie, Cunxiang Wang, Da Yin, Hao Zeng, Jiajie Zhang, et al. 2025. GLM-4.5: Agentic, reasoning, and coding (ARC) foundation models. *arXiv preprint arXiv:2508.06471*.